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Privacy-Preserving Non-negative Matrix Factorization for Decentralized-Data Using Correlated Noise

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Abstract

Several matrix factorization algorithms are employed in machine learning applications. Among these, Non-negative Matrix Factorization (NMF) gained attention due to the ability to extract meaningful features from inherently non-negative data, such as images or videos. However, such data are often privacy-sensitive, which necessitates formal privacy guarantees of the machine learning model training algorithm. Additionally, modern data are typically stored in different nodes or clients, rather than a centralized server. Conventional decentralized privacy-preserving schemes suffer from too much noise and consequently, much smaller utility compared to their centralized counterparts. This motivates us to develop an efficient privacy-preserving NMF algorithm that can operate on decentralized data, and closely approximates the performance of centralized/non-privacy-preserving approach, while offering strict privacy guarantees. We design our method and demonstrate our results in such a way that the clients/data holders have the control to select the degree of privacy guarantee based on the required utility. We show the effectiveness of our proposed algorithm on six real datasets. Our experimental results show that our proposed method easily outperforms conventional privacy-preserving scheme, while achieving close approximation of the non-privacy-preserving approach under some parameter choices.

Keywords: Non-negative Matrix Factorization (NMF), Differential Privacy (DP), Decentralized Data, Correlation Assisted Private Estimation (**CAPE**), Rényi Differential Privacy (RDP)

1 Introduction

Non-negative Matrix Factorization (NMF) is an algorithm that splits a matrix into two non-negative matrices for use in machine learning (ML) and data analysis [1]. Unlike other common matrix factorization algorithms, NMF applies constraints such that the component matrices are non-negative and comprehensible. More specifically, NMF solves the following problem:

$$\mathbf{V} \approx \mathbf{WH}, \quad (1)$$

where $\mathbf{V}$ is the data matrix with dimension $D \times N$, where N is the number of data points, and D is the dimension of each data point. The entries of $\mathbf{V}$ satisfy $v_{ij} \geq 0 \forall i, j$. The matrices $\mathbf{W}$ and $\mathbf{H}$ are often called *the basis or dictionary matrix* and *the coefficient or loading matrix*, respectively. The non-negativity property of the component matrices makes NMF suitable for tasks, such as feature extraction, pattern identification, and clustering, in a variety of applications, including but not limited to image processing and image segmentation, text mining, representation learning, community detection in social networks, and bio-informatics [2]. More specifically, NMF is a useful tool for gleaning important insights from complicated datasets of inherently non-negative data, and provide insightful features for ML models.

Privacy-sensitive learning is important in many signal processing and ML applications, such as human health research, business informatics, and location-based services. Releasing any function of private data, even summary statistics and other aggregates, can reveal information about the underlying training data. Differential privacy (DP) [3, 4] is a cryptographically motivated and mathematically rigorous framework for measuring the risk associated with performing computations on private data. More specifically, it measures the privacy risk in terms of the probability of identifying the presence of individual data points in a dataset from the results of computations performed on that data. As such, it has emerged as a de-facto standard for privacy-preserving technologies in research and practice [5–7]. DP is also useful when the private data is distributed over multiple locations (clients or nodes) and each client has its own privacy-sensitive dataset [8]. In such cases, the clients may be reluctant or unable to directly share “raw” data to an aggregator due to privacy and/or technical reasons. Decentralized data processing can allow these clients to maintain local control of the data, while sending functions of local data to an aggregator for a joint computation. If these computations are differentially private, they can measure and control privacy risks.

NMF and Privacy. As mentioned before, NMF offers several advantages over other matrix factorization algorithms for applications with inherently non-negative data. Such data are often privacy-sensitive, and the ML model parameters (as a matrix $\mathbf{W}$ or vector $\mathbf{w}$) that are learned by training on the users’ sensitive data, can cause privacy leakage. For instance, several works showed [9–11] that Membership Inference Attacks can identify users in the training set given the learned parameters. Other works [12–14] showed how personal data leakage occurred from modern ML and signal processing tasks. In addition, distributed ML systems that typically employ gradient sharing

among nodes, are also susceptible to leakage of private training information [15]. As such, differential privacy has been applied in distributed systems to preserve user-privacy.

However, differentially private algorithms introduce noise to guarantee privacy: conventional distributed DP algorithms often have poor utility due to excess noise compared to centralized analyses [8]. In this paper, we propose a differentially private NMF algorithm (**capeNMF**) for decentralized-data settings. We employ the Correlation Assisted Private Estimation (CAPE) framework [8], which is a novel privacy-preserving decentralized protocol that provides utility close to the centralized case. Recognizing the difficulty of privacy accounting for multi-round algorithms, we employ Rényi Differential Privacy [16] to provide a tight accounting of the overall privacy budget. We demonstrate that our proposed algorithm can achieve performance very close to non-private NMF on diverse datasets, while providing strict privacy guarantee.

1.1 Related Works

Most existing NMF algorithms obtain the component matrices by performing the minimization: $\min_{\mathbf{W} \in \mathcal{W}, \mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{WH}\|_F^2$, where $\mathcal{W} \subseteq \mathbb{R}_+^{D \times K}$ is the constraint set for $\mathbf{W}$. A number of techniques, including the multiplicative updates [17], alternating direction method of multipliers (ADMM) [18], block principal pivoting [19], active set method [20], and projected gradient descent (PGD) [21], have been developed to find the optimal point of this optimization problem. The majority of these algorithms rely on updating $\mathbf{W}$ and $\mathbf{H}$ in an alternating fashion. Our work is inspired by the robust NMF algorithm [22] that employed PGD. We refer the reader to the works of Wang and Zhang [23], Kim et al. [24] for a detailed analysis of these optimization strategies.

One of the most widely-used methods for ensuring that an algorithm satisfies differential privacy is to introduce randomization into the gradient calculation [25, 26]. Other common methods include the output perturbation [27], objective perturbation [28], the exponential mechanism [29], the Laplace mechanism [3], and smooth sensitivity [30]. Note that, depending on the application, different mechanisms may perform differently. Considering NMF, Saha and Imtiaz [2] have proposed a differentially private NMF algorithm for centralized-data settings, and performed theoretical analysis on the sensitivity and the privacy budget. To the best of our knowledge, no work has employed differential privacy in the generalized NMF decomposition for decentralized-data systems, and performed tight accounting for the privacy budget.

We note that, generalized optimization problems in decentralized-data settings have been investigated in numerous works. Boyd et al. [31], Molzahn et al. [32], Uribe et al. [33], Nedic and Ozdaglar [34], Tsitsiklis et al. [35], Huang et al. [36], Han et al. [37], Nozari et al. [38], Zhu et al. [39] showed different methods of optimization in decentralized-data settings. Song et al. [40] and Bassily et al. [41] employed randomized gradient calculations for guaranteeing differential privacy in optimization problems. Another commonly used method is the output perturbation [42], in which is often difficult to characterize the relationship between the parameter set and the data. This is because the optimization objective is often complicated, making it difficult to compute the sensitivity. Nevertheless, this can be avoided with differentially-private gradient descent by clipping the gradients, but at the cost of slowing down the convergence.

Another mechanism is the objective perturbation [38, 42], where the objective function is perturbed. Smith [43] presented a broad method for computing summary statistics, in addition to optimization problems, using the sample-and-aggregate framework, and both the Laplace and exponential mechanisms [44].

1.2 Our Contributions

As mentioned before, existing differentially-private algorithms for decentralized-data settings suffer from too much noise, and much lower utility compared to their centralized counterparts. Our goal in this work is to address this problem by developing an efficient differentially-private NMF algorithm (**capeNMF**) that can operate on privacy-sensitive non-negative data in a decentralized-data setting. We focus on releasing the basis/dictionary matrix $\mathbf{W}$. Since an adversary can obtain private information about users from the estimated $\mathbf{W}$, we employ the CAPE protocol [8] to ensure that the released $\mathbf{W}$ satisfies differential privacy guarantee. We design our algorithm in such a way that the estimated privacy-preserving $\mathbf{W}$ closely approximates the non-private $\mathbf{W}$ to be useful in practice. We summarize our contributions below:

- We develop an efficient privacy-preserving non-negative matrix factorization algorithm, capable of operating on privacy-sensitive data in a decentralized-data setting, that closely approximates the output of the non-private approach.
- We employ the CAPE protocol [8] to achieve the same performance as the centralized-data scenario in a symmetric decentralized-data setting, while guaranteeing differential privacy.
- We analyze our multi-round algorithm with the Rényi Differential Privacy [16], which provides a significantly better accounting of the overall privacy loss compared to the traditional strong composition rule [4].
- To demonstrate the efficacy of our suggested algorithm, we performed experiments on six real datasets. Our experiments clearly indicate that for certain parameter selections, our proposed approach can provide a close approximation to the non-private results, easily outperforming the conventional differentially private decentralized NMF.

2 Problem Formulation

2.1 Notations

For vector, matrix, and scalar, we used bold lower case letter ($\mathbf{v}$), bold capital letter ($\mathbf{V}$), and unbolded letter (M), respectively. We used subscript t to denote the iteration instant – $\mathbf{W}_t$, for instance, represents the dictionary matrix following t iterations. We indicate a single update by the superscript ‘+’. We used the symbol $\mathbf{v}_n$ for the n -th column of the matrix $\mathbf{V}$; and v_{ij} denotes the (i, j) -th entry of matrix $\mathbf{V}$. We sometimes denote the set $\{1, 2, \dots, N\}$ as $[N]$. Inequality $\mathbf{V} \geq 0$ apply entry-wise. We denoted $\mathcal{L}_2$ norm (Euclidean norm) with $\|\cdot\|_2$, the $\mathcal{L}_{1,1}$ norm with the $\|\cdot\|_{1,1}$, and the Frobenius norm with $\|\cdot\|_F$. $\mathbb{R}$, and $\mathbb{R}_+$ denote the set of real numbers, and set of positive real numbers, respectively. $\mathcal{P}_+$ denotes the euclidean projector, which projects

Notation	Meaning
$\mathbf{V}$	Data matrix
$\mathbf{W}$	Dictionary/basis matrix
$\mathbf{W}_{private}$	Differentially private $\mathbf{W}$
$\mathbf{H}$	Coefficient matrix
D	Ambient dimension of the matrix
K	Latent dimension
N	Number of user in the data matrix
P	Total number of clients
$\mathbf{v}_n$	n -th sample in data matrix
(ϵ, δ)	Privacy parameters
Δ	$\mathcal{L}_2$ Sensitivity
$\nabla f(\mathbf{W})$	Gradient of the loss function to update dictionary matrix
$\bar{\nabla} f(\mathbf{W})$	Noise perturbed gradient to learn $\mathbf{W}_{private}$
$\mathbf{A}, \mathbf{B}$	Statistics matrices of $\nabla f(\mathbf{W})$
$\bar{\mathbf{A}}, \bar{\mathbf{B}}$	Noise perturbed statistics matrices $\mathbf{A}, \mathbf{B}$

Table 1: Common notations used in the paper

onto the non-negative orthant. Lastly, we denoted the probability distribution function of zero mean unit variance Gaussian random variable as: $f(x) = \frac{1}{\sqrt{2\pi}} \exp(\frac{-x^2}{2})$. We summarized the frequently used notations at Table 1.

2.2 The NMF Problem

The two-block coordinate descent approach presented in Algorithm 1 is followed by the majority of the algorithms covered in Section 1. To decompose the data matrix $\mathbf{V}$ as $\mathbf{V} \approx \mathbf{WH}$, where $\mathbf{W} \in \mathcal{W}$ with $\mathcal{W} \subseteq \mathbb{R}_+^{D \times K}$, and $\mathbf{H} \in \mathbb{R}_+^{K \times N}$. When the dictionary matrix $\mathbf{W}$ is updated, the coefficient matrix $\mathbf{H}$ is kept constant. Subsequently, when the coefficient matrix $\mathbf{H}$ is updated, the revised dictionary matrix is kept constant. This process is continued until certain convergence conditions are satisfied. Several optimization algorithms are employed in such fashion; for example, the PGD, and ADMM [22]. The optimization problem of NMF can be formulated as:

$$\min_{\mathbf{W} \in \mathcal{W}, \mathbf{H} \geq 0} \left(\frac{1}{2} \|\mathbf{V} - \mathbf{WH}\|_F^2 \right). \quad (2)$$

Note that the NMF algorithm cannot guarantee an exact recovery of the original data matrix $\mathbf{V}$, as the objective function in equation 2 is not convex in both $\mathbf{W}$, and $\mathbf{H}$. Now, following the two-block coordinate descent method mentioned in the Algorithm 1, one can solve the optimization by i) updating the basis matrix based on the fixed coefficient matrix as $\mathbf{W}_{t+1} = \arg \min_{\mathbf{W} \in \mathcal{W}} L(\mathbf{V}, \mathbf{W}, \mathbf{H}_t)$; and ii) updating the coefficient matrix based on the latest dictionary matrix as $\mathbf{H}_{t+1} = \arg \min_{\mathbf{H} \geq 0} L(\mathbf{V}, \mathbf{W}_{t+1}, \mathbf{H})$. Here, the loss function is defined as $L(\mathbf{V}, \mathbf{W}, \mathbf{H}) = \frac{1}{2} \|\mathbf{V} - \mathbf{WH}\|_F^2$. Note that, a single step of updating $\mathbf{W}$ can be expressed [2] as $\mathbf{W}^+ = \mathcal{P}_{\mathcal{W}}(\mathbf{W} - \eta_{\mathcal{W}} \nabla f_{\mathbf{W}}(\mathbf{W}))$, where $f_{\mathbf{W}}(\mathbf{W}) = \frac{1}{2} \|\mathbf{V} - \mathbf{WH}_t\|_F^2 = \frac{1}{2} \text{tr}(\mathbf{W}^\top \mathbf{W} \mathbf{A}_t) - \text{tr}(\mathbf{W}^\top \mathbf{B}_t)$. We define two matrices: $\mathbf{A}_t = \frac{1}{N} \mathbf{H}_t \mathbf{H}_t^\top$, and $\mathbf{B}_t = \frac{1}{N} \mathbf{V} \mathbf{H}_t^\top$, for the brevity and convenience during the inclusion of differential privacy, as shown later. The gradient of $f_{\mathbf{W}}(\mathbf{W})$ with respect to $\mathbf{W}$

Algorithm 1 Generalized NMF Algorithm: Two-Block Coordinate Descent Method [45]

Require: Data matrix $\mathbf{V}$, number of iterations T

Ensure: Dictionary matrix $\mathbf{W}$

- 1: Initialization: $\mathbf{W}_0 \in \mathcal{W}$, $\mathbf{H}_0 \in \mathbb{R}_+^{K \times N}$
 - 2: **for** $t = 0$ to $T - 1$ **do**
 - 3: $\mathbf{W}_{t+1} \leftarrow \text{update}(\mathbf{V}, \mathbf{W}_t, \mathbf{H}_t)$
 - 4: $\mathbf{H}_{t+1} \leftarrow \text{update}(\mathbf{V}, \mathbf{W}_{t+1}, \mathbf{H}_t)$
 - 5: **end for**
 - 6: **return** Optimized dictionary matrix $\mathbf{W}$
-

is $\nabla f_{\mathbf{W}}(\mathbf{W}) = \mathbf{W}\mathbf{A}_t - \mathbf{B}_t$. According to the PGD approach [2, 22], the constrained projection function keeps the columns of $\mathbf{W}$ in the unit $\mathcal{L}_2$ ball. That is, each column is updated as:

$$\mathbf{w}_k^+ := \frac{\mathcal{P}_+(\mathbf{w}_k - \eta_W \nabla f_W(\mathbf{w}_k))}{\max\left(1, \|\mathcal{P}_+(\mathbf{w}_k - \eta_W \nabla f_W(\mathbf{w}_k))\|_2\right)}, \forall k \in [K]. \quad (3)$$

Similarly, a single step of updating $\mathbf{H}$ can be expressed [2] as $\mathbf{H}^+ = \mathcal{P}_+(\mathbf{H} - \eta_H \nabla f_{\mathbf{H}}(\mathbf{H}))$, where $f_{\mathbf{H}}(\mathbf{H}) = \frac{1}{2} \|\mathbf{V} - \mathbf{W}_{t+1}\mathbf{H}\|_F^2$, and $\nabla f_{\mathbf{H}}(\mathbf{H}) = \frac{1}{N}(\mathbf{W}_{t+1}^\top \mathbf{W}_{t+1}\mathbf{H} - \mathbf{W}_{t+1}^\top \mathbf{V})$.

3 Proposed Method

In this section, we describe our proposed algorithm to estimate $\mathbf{W}_{private}$ in a decentralized-data setting. We specify our privacy model, and analytically show how to incorporate differential privacy for estimating the basis matrix $\mathbf{W}_{private}$, such that it provides close approximation to the true basis matrix $\mathbf{W}$. Note that, $\mathbf{H}$ is not released to the public. Next, we analyze our multi-round gradient descent algorithm with RDP [16] to attain a tight characterization of the privacy loss encountered over the course of the optimization process. And finally, we discuss regarding the convergence of the algorithm. In Appendix A, we reviewed the necessary definitions and propositions according to Saha and Intiaz [2] for describing our proposed approach.

Trust/Collusion Model. As mentioned before, we employ the CAPE scheme [8] in our proposed algorithm. We assume that all of the P clients/sites and the central aggregator follow the protocol honestly (see Figure 1). However, up to $P_C = \lceil \frac{P}{3} \rceil - 1$ sites can collude with an adversary to learn about some client's data/function output. The central aggregator is also honest-but-curious (and therefore, can collude with an adversary). An adversary can observe the outputs from each site, as well as the output from the aggregator. Additionally, the adversary can know everything about the colluding sites (including their private data). We denote the number of non-colluding sites with P_H such that $P = P_C + P_H$. Without loss of generality, we designate the non-colluding sites with $\{1, \dots, P_H\}$.

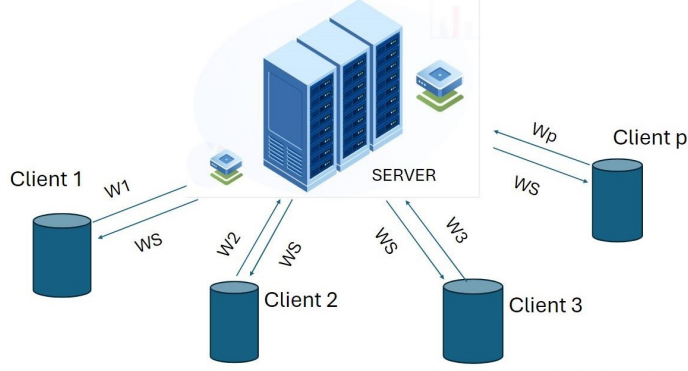


Fig. 1: Schematic diagram of the decentralized-data setting for the proposed capeNMF algorithm.

In our decentralized-data setting, “processing” takes place in the clients and the central aggregator. More specifically, each client p hosts N_p data samples $\mathbf{V}_p \in \mathbb{R}^{D \times N_p}$ for $p \in [P]$. The total number of data samples across all the clients is $N = \sum_{p=1}^P N_p$. During the decentralized optimization, the clients update and host the local coefficient matrices $\mathbf{H}_p \in \mathbb{R}^{K \times N_p}$. The global basis matrix is updated in the central aggregator. To this end, the clients compute the gradient $\nabla f_{\mathbf{W}}(\mathbf{W})$ on the local privacy-sensitive data. Since the aggregator can collude with an adversary, the computation of the gradients have to satisfy DP guarantee in order for ensuring the privacy of the global $\mathbf{W}$. After each update, the central aggregator sends the most recent $\mathbf{W}$ to all the clients.

As mentioned in Saha and Imtiaz [2], the clients can choose from several mechanisms to ensure DP: Laplace mechanism, Gaussian mechanism, Exponential mechanism [4, 29]. We employ the Gaussian mechanism for this work in accordance with Imtiaz et al. [8]. More specifically, we employ a variant of the input perturbation: the noisy gradient descent, where we employ the CAPE scheme to estimate the gradient $\nabla f_{\mathbf{W}}(\mathbf{W})$ satisfying DP [25, 26].

Estimating the Differentially-private $\mathbf{W}$ in the Decentralized-data Setting.

Our proposed capeNMF algorithm employs the CAPE protocol [8] to perform the privacy-preserving iterative message-passing between the clients and the aggregator to compute the DP basis matrix $\mathbf{W}$. As mentioned before, the central aggregator initializes $\mathbf{W}_0$, and sends it to all the clients. Each client p updates their local coefficient matrix based on the received $\mathbf{W}_0$ as

$$\mathbf{H}_{p,t+1} = \arg \min_{\mathbf{H} \geq 0} L(\mathbf{V}_p, \mathbf{W}_t, \mathbf{H}),$$

for iteration $t \geq 0$, where $L(\mathbf{V}_p, \mathbf{W}_t, \mathbf{H}) = \frac{1}{2} \|\mathbf{V}_p - \mathbf{W}_t \mathbf{H}\|_F^2$. This optimization process does not need to satisfy DP, since the clients do not send the updated $\mathbf{H}_p$ anywhere. Once the optimization is completed (that is, after certain stopping condition(s) are

met), client p computes the following two matrices locally:

$$\begin{aligned}\mathbf{A}_{p,t+1} &= \frac{1}{N_p} \mathbf{H}_{p,t+1} \mathbf{H}_{p,t+1}^\top \\ \mathbf{B}_{p,t+1} &= \frac{1}{N_p} \mathbf{V}_p \mathbf{H}_{p,t+1}^\top.\end{aligned}$$

Note that, since $\mathbf{H}_p$ is a function of the privacy-sensitive data $\mathbf{V}_p$, both $\mathbf{A}_{p,t+1}$ and $\mathbf{B}_{p,t+1}$ are privacy-sensitive. Therefore, to ensure DP guarantee, the clients need to send noise perturbed versions of these matrices to the aggregator according to CAPE, such that the aggregator can compute the gradient $\nabla f_{\mathbf{W}}(\mathbf{W})$. To that end, the $\mathcal{L}_2$ sensitivity of these two matrices are [2] given by

$$\begin{aligned}\Delta_{\mathbf{A}_p} &= \frac{2}{N_p} \times (\max \mathcal{L}_2 \text{ norm of columns of } \mathbf{H}_p)^2 \\ \Delta_{\mathbf{B}_p} &= \frac{2}{N_p} \times (\max \mathcal{L}_2 \text{ norm of columns of } \mathbf{V}_p).\end{aligned}$$

Note that, due to our normalization step (see Steps 1 and 5 in Algorithm 3), these quantities are not dependent on iteration t , and therefore, we removed the subscript. Now, according to the CAPE protocol [8], each client p computes the following differentially private matrices:

$$\overline{\mathbf{A}_{p,t+1}} = \mathbf{A}_{p,t+1} + \mathbf{E}_{p,t+1}^A + \mathbf{K}_{p,t+1}^A \quad (4)$$

$$\overline{\mathbf{B}_{p,t+1}} = \mathbf{B}_{p,t+1} + \mathbf{E}_{p,t+1}^B + \mathbf{K}_{p,t+1}^B, \quad (5)$$

where $\mathbf{E}_{p,t+1}^A$ and $\mathbf{E}_{p,t+1}^B$ are generated according to the zero-sum noise generation algorithm presented in Algorithm 2 [8]. Additionally, each client p generates the following two noise terms independently and locally:

- $\mathbf{K}_{p,t+1}^A \in \mathbb{R}^{K \times K}$ with i.i.d. entries from $\mathcal{N}(0, \tau_{AK}^2)$, where $\tau_{AK}^2 = \frac{1}{P} \tau_A^2$
- $\mathbf{K}_{p,t+1}^B \in \mathbb{R}^{D \times K}$ with i.i.d. entries from $\mathcal{N}(0, \tau_{BK}^2)$, where $\tau_{BK}^2 = \frac{1}{P} \tau_B^2$.

Here, we note that according to the Gaussian mechanism [4], computing (ϵ, δ) DP approximates of $\mathbf{A}_p$ and $\mathbf{B}_p$ requires noise standard deviations $\tau_{A,p}$ and $\tau_{B,p}$ to satisfy

$$\tau_{A,p} = \frac{\Delta_{\mathbf{A}_p}}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}, \quad (6)$$

$$\tau_{B,p} = \frac{\Delta_{\mathbf{B}_p}}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}. \quad (7)$$

For simplicity, we consider the symmetric case, i.e., $N_p = \frac{N}{P}$, which leads to $\Delta_{\mathbf{A}_p} = \Delta_{\mathbf{A}}$, and $\Delta_{\mathbf{B}_p} = \Delta_{\mathbf{B}}$ for all p . Consequently, $\tau_{A,p} = \tau_A = \frac{\Delta_{\mathbf{A}}}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}$, and $\tau_{B,p} = \tau_B = \frac{\Delta_{\mathbf{B}}}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}$.

Algorithm 2 Generate zero-sum noise [8]

Require: Local noise variances $\{\tau_p^2\}$

Ensure: Zero-sum noise terms $\{e_p\}$

- 1: Each site generates $\hat{e}_p \sim \mathcal{N}(0, \tau_p^2)$
 - 2: Aggregator computes $\sum_{p=1}^P \hat{e}_p$ using a non-colluding party (Secure MPC) [46]
 - 3: Aggregator broadcasts $\sum_{p=1}^P \hat{e}_p$ to all sites $p \in [P]$
 - 4: Each site computes $e_p = \hat{e}_p - \frac{1}{P} \sum_{p'=1}^P \hat{e}_{p'}$
 - 5: **return** e_p
-

At each iteration t , the clients compute the noisy estimates of $\mathbf{A}_{p,t+1}$ and $\mathbf{B}_{p,t+1}$ as shown in equation 4 and equation 5. These two terms are then sent to the aggregator and the aggregator computes:

$$\overline{\mathbf{A}}_{t+1} = \frac{1}{P} \sum_{p=1}^P \overline{\mathbf{A}}_{p,t+1} \quad (8)$$

$$\overline{\mathbf{B}}_{t+1} = \frac{1}{P} \sum_{p=1}^P \overline{\mathbf{B}}_{p,t+1}. \quad (9)$$

These matrices are then used by the central aggregator to compute the global loss function $f_{\mathbf{W}}(\mathbf{W}) = \frac{1}{2} \text{tr}(\mathbf{W}^\top \mathbf{W} \overline{\mathbf{A}}_{t+1}) - \text{tr}(\mathbf{W}^\top \overline{\mathbf{B}}_{t+1})$. Note that, for updating the global $\mathbf{W}$, the gradient of this function is needed, which is given by $\nabla f_{\mathbf{W}}(\mathbf{W}) = \mathbf{W}_t \overline{\mathbf{A}}_{t+1} - \overline{\mathbf{B}}_{t+1}$. By Lemma 1 of Imtiaz et al. [8], the variances of the noise of the two estimates: $\overline{\mathbf{A}}_{t+1}$ and $\overline{\mathbf{B}}_{t+1}$, are exactly the same as the pooled-data scenario in the symmetric setting. Consequently, the gradient estimate $\nabla f_{\mathbf{W}}(\mathbf{W})$ is also the same as the pooled-data scenario. The complete algorithm is shown in Algorithm 3.

Theorem 1 (Privacy of Algorithm 3). *Consider Algorithm 3 in a decentralized-data setting described in Section 3 with $N_p = N/P$ for all $p \in [P]$. Then Algorithm 3 releases a $\left(\frac{P^2 T \alpha_{\text{opt}}}{2} \left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2} \right) + \frac{\log \frac{1}{\delta}}{\alpha_{\text{opt}} - 1}, \delta \right)$ differentially-private basis matrix $\mathbf{W}_{\text{private}}$ for any $0 < \delta < 1$ after T iterations, where $\alpha_{\text{opt}} = 1 + \sqrt{\frac{2}{P^2 T \left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2} \right)} \log \frac{1}{\delta}}$.*

Proof. To prove Theorem 1, we analyze the **capeNMF** algorithm with Rényi Differential Privacy [16]. As mentioned before, analyzing the total privacy loss of a multi-shot algorithm, each stage of which satisfies DP, is a challenging task. It has been shown [16, 47] that the advanced composition theorem [4] for (ϵ, δ) -differential privacy can be loose. The main reason is that one can formulate infinitely many (ϵ, δ) -DP algorithms for a given noise variance τ^2 [8]. RDP offers a much simpler composition rule that is shown to be tight. Recall that at each iteration t of **capeNMF**, we compute the noise perturbed DP estimates of the two matrices $\mathbf{A}_{t+1}$ and $\mathbf{B}_{t+1}$. As we employed the CAPE scheme in the symmetric setting, the variances of noise at the aggregator for $\overline{\mathbf{A}}_{t+1}$ and $\overline{\mathbf{B}}_{t+1}$ are: $\sigma_{\mathbf{A}}^2 = \frac{\tau_A^2}{P^2}$ and $\sigma_{\mathbf{B}}^2 = \frac{\tau_B^2}{P^2}$, respectively. From Proposition 3, we have that the

Algorithm 3 Differentially-private NMF for Decentralized-data Settings (capeNMF)

Require: Data $\{\mathbf{V}_p \in \mathbb{R}^{D \times N_p} : p \in [P]\}$, step size η_H and η_W , maximum number of iterations T , local noise standard deviations $\{\tau_A, \tau_B\}$, local sensitivities $\{\Delta_A, \Delta_B\}$.

Ensure: Differentially private dictionary matrix $\mathbf{W}_{\text{private}}$

- 1: Initialize: $\mathbf{W}_0$, pre-process data to ensure $\|\mathbf{v}_{p,n}\|_2 \leq 1$ for all $n \in [N_p]$ and $p \in [P]$.
 - 2: **for** $t = 0$ to $T - 1$ **do**
 - 3: **for all** clients $p = 1, 2, \dots, P$ **do**
 - 4: Solve $\mathbf{H}_{p,t+1} = \arg \min_{\mathbf{H} \geq 0} L(\mathbf{V}_p, \mathbf{W}_t, \mathbf{H}) = \arg \min_{\mathbf{H} \geq 0} \frac{1}{2} \|\mathbf{V}_p - \mathbf{W}_t \mathbf{H}\|_F^2$
 - 5: Ensure $\|\mathbf{h}_{p,n}\|_2 \leq 1$ for all $n \in [N_p]$ using $\mathbf{h}_{p,n} = \frac{\mathbf{h}_{p,n}}{\max(1, \|\mathbf{h}_{p,n}\|_2)}$
 - 6: Generate $\mathbf{E}_{p,t+1}^A \in \mathbb{R}^{K \times K}$; $\mathbf{E}_{p,t+1}^B \in \mathbb{R}^{D \times K}$ using Algo. 2 (element-wise)
 - 7: Generate $\mathbf{K}_{p,t+1}^A \in \mathbb{R}^{K \times K}$ and $\mathbf{K}_{p,t+1}^B \in \mathbb{R}^{D \times K}$, as described in the text
 - 8: Compute $\mathbf{A}_{p,t+1} = \frac{1}{N_p} \mathbf{H}_{p,t+1} \mathbf{H}_{p,t+1}^\top$, and $\mathbf{B}_{p,t+1} = \frac{1}{N_p} \mathbf{V}_p \mathbf{H}_{p,t+1}^\top$
 - 9: Set $\overline{\mathbf{A}}_{p,t+1} = \mathbf{A}_{p,t+1} + \mathbf{E}_{p,t+1}^A$; $\overline{\mathbf{B}}_{p,t+1} = \mathbf{B}_{p,t+1} + \mathbf{E}_{p,t+1}^B$
 - 10: Send $\overline{\mathbf{A}}_{p,t+1}$ and $\overline{\mathbf{B}}_{p,t+1}$ to the aggregator
 - 11: **end for**
 - 12: Compute the matrices $\overline{\mathbf{A}}_{t+1} = \frac{1}{P} \sum_{p=1}^P \overline{\mathbf{A}}_{p,t+1}$, and $\overline{\mathbf{B}}_{t+1} = \frac{1}{P} \sum_{p=1}^P \overline{\mathbf{B}}_{p,t+1}$
 - 13: Compute updated loss function $f_{\mathbf{W}}(\mathbf{W}) = \frac{1}{2} \text{tr}(\mathbf{W}^\top \mathbf{W} \overline{\mathbf{A}}_{t+1}) - \text{tr}(\mathbf{W}^\top \overline{\mathbf{B}}_{t+1})$
 - 14: Solve the optimization problem $\mathbf{W}_{t+1} = \arg \min_{\mathbf{W} \in \mathcal{W}} f_{\mathbf{W}}(\mathbf{W})$
 - 15: Send $\mathbf{W}_{t+1}$ to all clients
 - 16: **end for**
 - 17: **return** $\mathbf{W}_{\text{private}} = \mathbf{W}_T$
-

computation of $\overline{\mathbf{A}}_{t+1}$ and $\overline{\mathbf{B}}_{t+1}$ satisfy $\left(\alpha, \frac{\alpha}{2\left(\frac{\tau_A}{P\Delta_A}\right)^2}\right)$ -RDP and $\left(\alpha, \frac{\alpha}{2\left(\frac{\tau_B}{P\Delta_B}\right)^2}\right)$ -RDP, respectively. According to Proposition 2, each step of Algorithm 3 is $\left(\alpha, \frac{P^2\alpha}{2}\left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2}\right)\right)$ -RDP. If the number of required iterations for reaching convergence is T , then under T -fold composition of RDP, the overall algorithm is $\left(\alpha, \frac{P^2T\alpha}{2}\left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2}\right)\right)$ -RDP. From Proposition 1, we have that the algorithm satisfies $\left(\frac{P^2T\alpha}{2}\left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2}\right) + \frac{\log \frac{1}{\delta}}{\alpha-1}, \delta\right)$ -DP for any $0 < \delta < 1$. For a given δ , we can compute the optimal (that provides the smallest overall ϵ) α as $\alpha_{\text{opt}} = 1 + \sqrt{\frac{2}{P^2T\left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2}\right)} \log \frac{1}{\delta}}$. This α_{opt} provides the lowest privacy risk. Therefore, Algorithm 3 releases a $\left(\frac{P^2T\alpha_{\text{opt}}}{2}\left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2}\right) + \frac{\log \frac{1}{\delta}}{\alpha_{\text{opt}}-1}, \delta\right)$ differentially-private basis matrix $\mathbf{W}_{\text{private}}$ for any $0 < \delta < 1$. $\square$

Remark 1 (Convergence and Norm Clipping). *As described in Saha and Imtiaz [2], we note that the objective function is non-increasing under the two update steps (i.e., Steps 4 and 14 of Algorithm 3), and the objective function is bounded below. Additionally, the noisy estimates of $\mathbf{A}$ and $\mathbf{B}$, and consequently the gradient of $f_{\mathbf{W}}(\mathbf{W})$ essentially contain zero mean noise. Although this does not provide guarantees on the*

Dataset	N	D	K
Guardian News Articles	4551	10285	8
UCI News Aggregator	3000	93	7
RCV1	9625	2999	4
TDT2	9394	3677	30
YaleB	2414	1024	38
CBCL	2429	361	50

Table 2: Summary of datasets

excess error, the estimate of the gradient converges in expectation to the true gradient [48]. Therefore, the optimization $\mathbf{W}_{t+1} = \arg \min_{\mathbf{W} \in \mathcal{W}} f_{\mathbf{W}}(\mathbf{W})$ should converge given sufficient steps and a small-enough step size. Additionally, we note that the norm clipping ensured by the normalizing steps in Algorithm 3 destroys the unbiasedness of the gradient estimate [47]. In general, clipping prescribes taking a smaller step “downhill” towards the optimal point [26] and may slow down the convergence.

4 Experimental Results

In this section, we demonstrate how effective our proposed **capeNMF** algorithm is by comparing it against a non-private NMF algorithm (**non – private**), and a differentially-private NMF algorithm employing conventional decentralized DP scheme (**conventional – DP**). We note that, according to Proposition 1 of Imtiaz et al. [8], our **capeNMF** algorithm should provide much better results than the **conventional – DP** algorithm. As in Saha and Imtiaz [2], we define the Objective Value in equation 10 as the performance index, which quantifies how well the algorithm can decompose the data matrix $\mathbf{V}$:

$$\text{Objective Value} = \frac{1}{2N} \|\mathbf{V} - \mathbf{WH}\|_F^2. \quad (10)$$

The reason for choosing this performance index is that we intend to evaluate how well the estimated basis and coefficient matrices *explain* the data $\mathbf{V}$. Another reason is that, we cannot use Mean Absolute Error (MAE) or Root-mean Squared Error (RMSE) with the estimated basis matrix [2]. This is because, often times we do not know the true basis matrix (although we can assume the basis matrix estimated from a non-private NMF algorithm is the true basis matrix). But more importantly, the order of the columns of the basis matrix is undefined. Therefore, we choose to use the Objective Value as defined above as the performance index.

In addition to a low *objective value*, we also want to achieve a low *overall* ϵ , since it represents the overall privacy budget of a multi-round algorithm. From Theorem 1, we note that the overall privacy budget of our proposed **capeNMF** algorithm is given by

$$\text{Overall } \epsilon = \frac{P^2 T \alpha_{\text{opt}}}{2} \left(\frac{\Delta_A^2}{\tau_A^2} + \frac{\Delta_B^2}{\tau_B^2} \right) + \frac{\log \frac{1}{\delta}}{\alpha_{\text{opt}} - 1},$$

for any target $\delta \in (0, 1)$. In our experiments, we have used the overall ϵ as another performance index.

We utilize six real datasets, as in Saha and Imtiaz [2], in order to assess our proposed **capeNMF** algorithm. Two of these datasets are face image datasets, and four are text datasets. In Table 2, we show the dataset sizes as well as the associated latent dimensions [2].

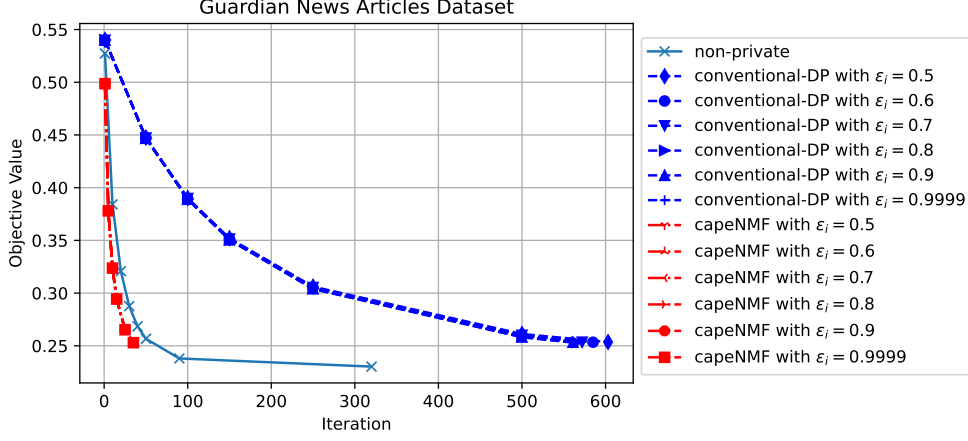
Hyperparameter selection. All hyper-parameter values, except for the learning rates, are used as described in Zhao and Tan [22]. For the **non-private** and **capeNMF** algorithms, we used an initial learning rate of 0.01 for all datasets. For the **conventional-DP** algorithm, we used an initial learning rate of 0.01 for all datasets, except the UCI News Aggregator dataset, and the TDT2 dataset, for which we used initial learning rates of 0.005 and 0.0005, respectively. As mentioned in Section 3, each client updates its local coefficient matrix $\mathbf{H}_p$ in a non-privacy-preserving way, but sends the noise perturbed $\mathbf{A}_p$ and $\mathbf{B}_p$ matrices to the aggregator for updating the global $\mathbf{W}$. Saha and Imtiaz [2] observed that the Non Negative Double Singular Value Decomposition (NNDSVD) [49] technique performs better than random initialization in minimizing the objective value, thus we used it to initialize $\mathbf{W}_0$.

We set maximum number of iterations $T = 10,000$, local noise standard deviations $\tau_A = \frac{\Delta_A}{\epsilon_i} \sqrt{2 \log \frac{1.25}{10^{-5}}}$ and $\tau_B = \frac{\Delta_B}{\epsilon_i} \sqrt{2 \log \frac{1.25}{10^{-5}}}$ for our experiments, where ϵ_i is the privacy parameter per iteration, Δ_A and Δ_B are the $\mathcal{L}_2$ sensitivities of $\mathbf{A}_p$ and $\mathbf{B}_p$, respectively, in the symmetric setting. Additionally, we used the number of clients $P = 10$, number of colluding clients $P_C = \lceil \frac{P}{3} \rceil - 1$ [8, 46], and the target $\delta = 10^{-5}$. We show the average performance over 10 independent runs. Note that, the choice of hyper-parameters is non-trivial [50] and corresponding end-to-end privacy analysis is still an open problem.

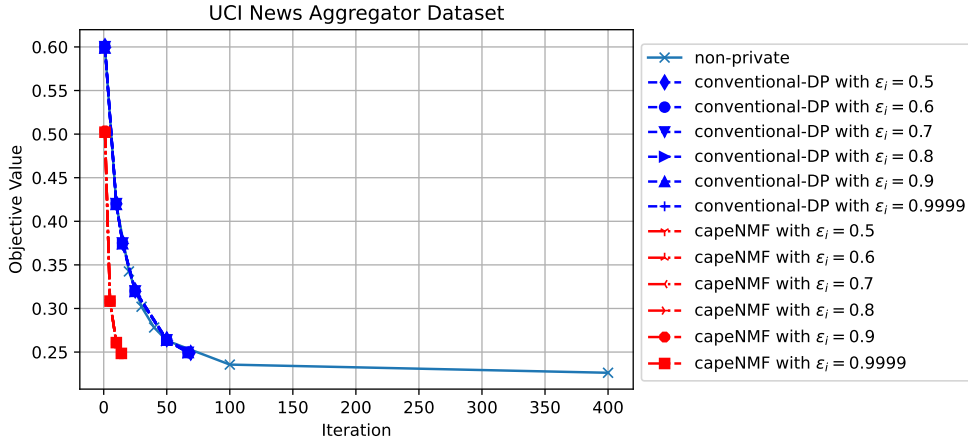
4.1 Experiments on Text Datasets

As mentioned before, we investigated the performance of our proposed **capeNMF** algorithm for topic modeling application on four text datasets. Note that, topic modeling aims to find abstract topics present in a collection of text documents. More specifically, topic modeling identifies distinct clusters of words that corresponds to a specific topic in the document [2]. The **Guardian News Articles Dataset** [51] has 4551 news articles, which are consolidated from the Guardian News API in 2006. For this dataset, we chose $K = 8$. The **UCI News Aggregator Dataset** [52] contains news collected between 10 March 2014 to 10 August 2014 from a web aggregator. This dataset contains 422937 news articles covering a variety of subjects. We randomly selected 750 articles from each category for our experiments. The **RCV1 Dataset** [53] is an archive of Reuters Corpus Volume I, which includes more than 800000 news wires that have been manually categorized. We choose about 1/10-th of the features at random and considered 9625 documents for our experiments. Finally, the **TDT2 Dataset** [54] contains 9394 records. In this case as well, we choose 1/10-th of the features at random.

Performance Comparison on Text Datasets. In Figures 2 and 3, we compare the **non-private**, **conventional-DP** and **capeNMF** algorithms in terms of *objective*



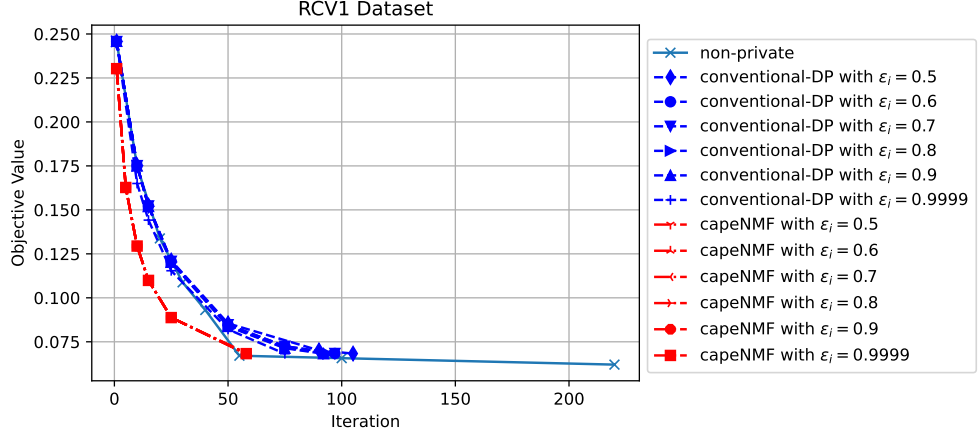
(a) Guardian News Articles



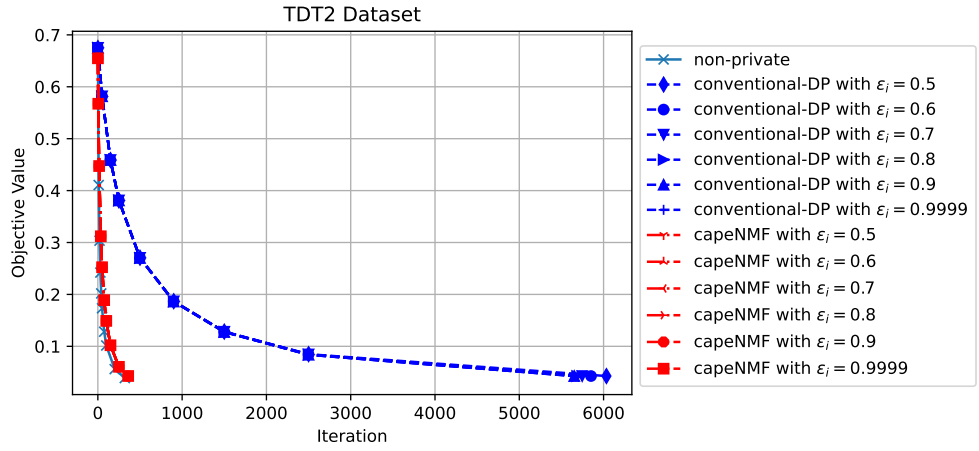
(b) UCI News Aggregator

Fig. 2: Variation of Objective Value with epoch for different ϵ_i on Guardian News Articles and UCI News Aggregator Datasets

value on the aforementioned text datasets. For the conventional – DP and capeNMF algorithms, we show how the objective value decreases with epochs for different ϵ_i values. It is evident from the plots that our proposed capeNMF algorithm performs very well, closely approximating the results of the non – private algorithm. The proposed capeNMF algorithm outperforms the conventional – DP algorithm by a substantial margin. Additionally, comparing the convergence speed, we observe that our proposed capeNMF algorithm needs lower number of epochs to reach the optimal point.



(a) RCV1



(b) TDT2

Fig. 3: Variation of Objective Value with epoch for different ϵ_i on RCV1 and TDT2 Datasets

Variation of Overall ϵ with ϵ_i on Text Datasets. In Figure 4, we show the variation of the final objective value (when the optimization is terminated), and the overall ϵ with ϵ_i for the conventional – DP algorithm and our proposed **capeNMF** algorithm on the text datasets. We also show the objective value of the non – private algorithm as a baseline. We observe from the plots that if enough iterations are allowed, both the conventional – DP and **capeNMF** algorithms can closely approximate the objective value of the non – private algorithm. However, since the overall ϵ depends on the number of iterations, and the proposed **capeNMF** algorithms typically requires a much smaller number of iterations to converge compared to conventional – DP, the

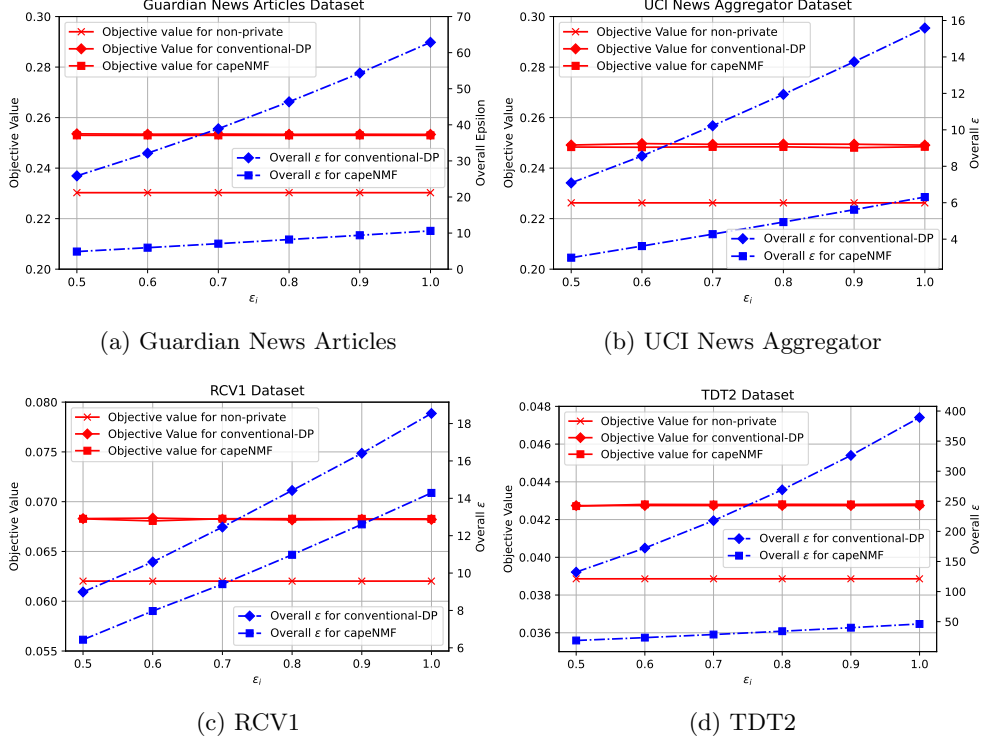


Fig. 4: Variation of Overall ϵ and Objective Value with ϵ_i on Text Datasets

overall ϵ achieved by **capeNMF** is much smaller than that of **conventional – DP** algorithm. That is, our proposed **capeNMF** algorithm can provide a much better privacy guarantee compared to the **conventional – DP** approach.

Variation of Objective Value with Dataset Fraction on Text Datasets.

In Figure 5, we show the variation of the final objective value with fraction of the dataset used for a fixed $\epsilon_i = 0.5$. We observe for all the text datasets that using the entire dataset produces the best results. In other words, our proposed algorithm performs better when more data samples are available. To explain this phenomena, we recall that the noise variance at the aggregator for $\overline{\mathbf{A}}_{t+1}$ and $\overline{\mathbf{B}}_{t+1}$ are: $\sigma_{\mathbf{A}}^2 = \frac{\tau_{\mathbf{A}}^2}{P^2} = \frac{\Delta_{\mathbf{A}}^2}{P^2 \epsilon_i^2} \times 2 \log \frac{1.25}{\delta} = \frac{8}{N^2 \epsilon_i^2} \log \frac{1.25}{\delta}$ and $\sigma_{\mathbf{B}}^2 = \frac{\tau_{\mathbf{B}}^2}{P^2} = \frac{\Delta_{\mathbf{B}}^2}{P^2 \epsilon_i^2} \times 2 \log \frac{1.25}{\delta} = \frac{8}{N^2 \epsilon_i^2} \log \frac{1.25}{\delta}$, respectively. It is evident that the noise variances $\sigma_{\mathbf{A}}^2$ and $\sigma_{\mathbf{B}}^2$ are inversely proportional with N^2 , where N is the total number of samples across all clients. Since $\overline{\mathbf{A}}_{t+1}$ and $\overline{\mathbf{B}}_{t+1}$ are instrumental in computing the gradient of the loss function, having more data samples results in better estimates of the gradient, which in turn achieved a better optimal point for our proposed **capeNMF**.

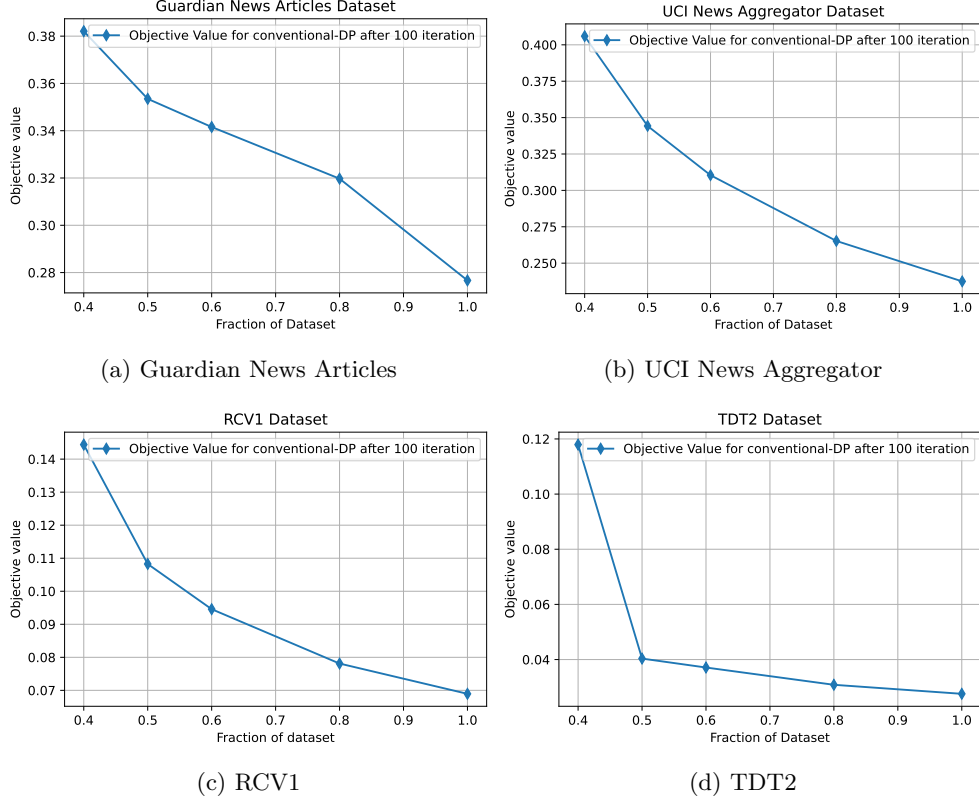


Fig. 5: Variation of Objective Value with dataset fraction on Text Datasets

Variation of Overall ϵ with Stopping Criteria on Text Datasets. We investigate the effect of stopping criteria on the overall ϵ of our proposed **capeNMF** algorithm. More specifically, assume that the aggregator stops the iteration when the following is satisfied: $|L_{\text{non-private}} - L_{\text{capeNMF}}| \leq \theta$, where $L_{\text{non-private}}$ and L_{capeNMF} are the final objective values attained by the non-private algorithm, and the proposed **capeNMF** algorithm, respectively. Intuitively, smaller θ would require bigger number of iterations, which would result in bigger overall ϵ for a fixed ϵ_i . In Figure 6, we observe that for all the text datasets, the more closely we want to approximate the non-private algorithm’s objective value, the more iteration we would need, which inevitably results in higher overall ϵ .

4.2 Experiments on Face Datasets

As mentioned before, we investigate the performance of our proposed **capeNMF** algorithm on two face datasets. The **YaleB Dataset** [55] contains 2414 face images of size 32×32 pixels, which are taken under various lighting circumstances. The dataset has 38 participants, including both male and female subjects. The **CBCL Dataset** [56]

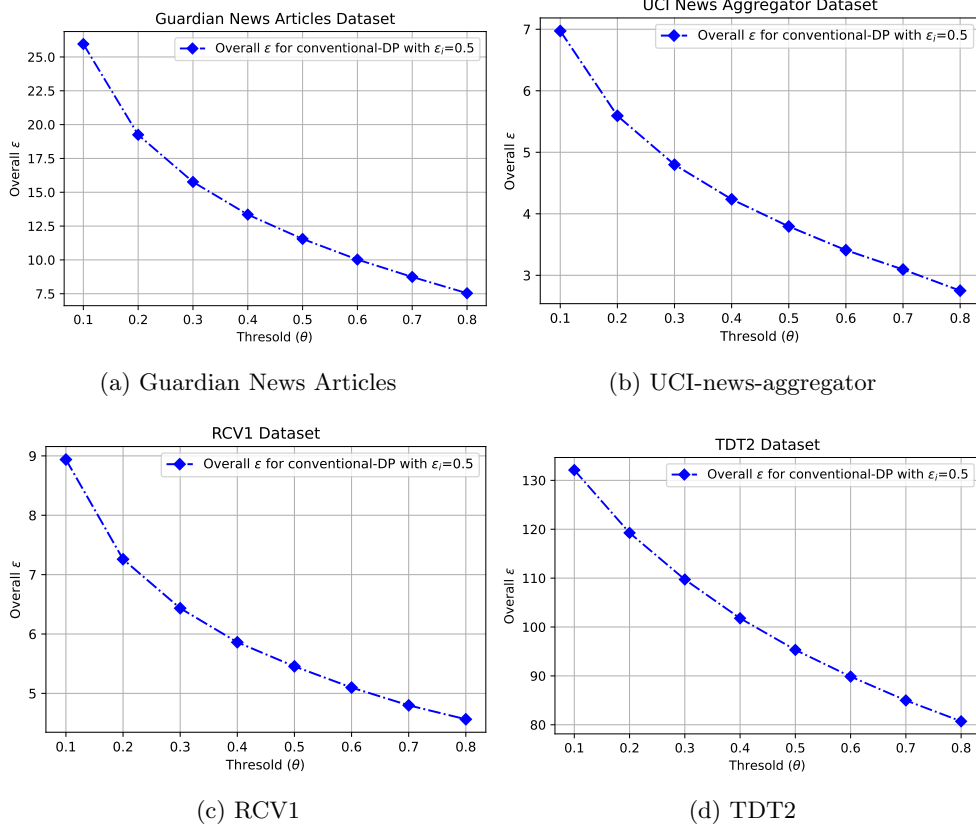
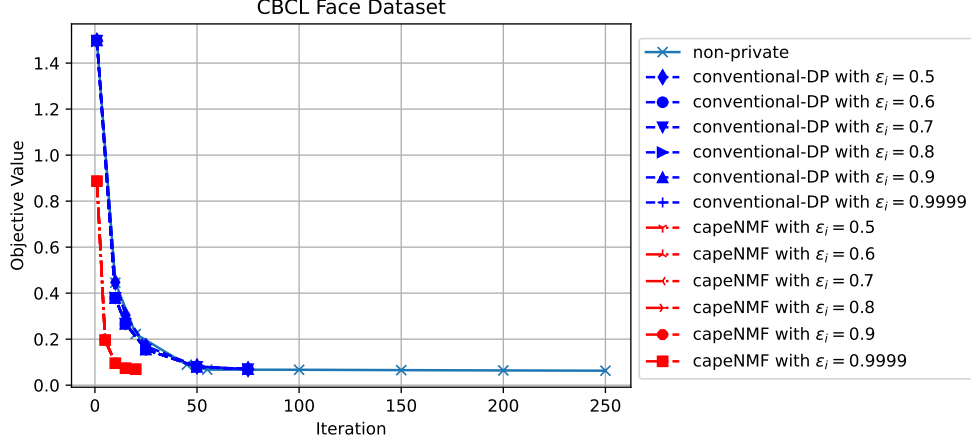


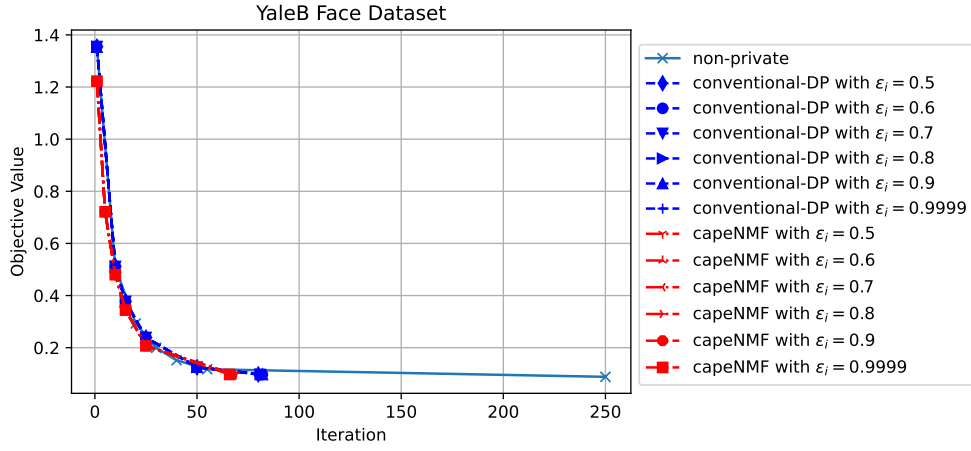
Fig. 6: Variation of Overall ϵ with Stopping Criteria on Text Datasets

contains 2429 face images of size 19×19 pixels. The hand-aligned frontal photographs are pre-processed by clipping to the range $[0, 1]$ after being linearly adjusted to equal the pixel mean and standard deviation of 0.25 [2]. We note that, the basis matrix $\mathbf{W}$ contributes towards *explaining* every face image in the corresponding dataset. As with the text datasets, we compare the performance of our proposed **capeNMF** with that of **non-private** and **conventional-DP** algorithms in terms of objective value and overall ϵ .

Performance Comparison on Face Datasets. In Figure 7, we compare the **non-private**, **conventional-DP** and **capeNMF** algorithms in terms of objective value on the face datasets. As with the text datasets, we show how the objective value decreases with epochs for different ϵ_i values for the **conventional-DP** and **capeNMF** algorithms. For both datasets, we observe that our proposed **capeNMF** algorithm performs very well – exceeding, if not closely approximating the results of the **non-private** algorithm. The proposed **capeNMF** algorithm outperforms the **conventional-DP** algorithm, albeit by a smaller margin compared to the text datasets. Additionally,



(a) CBCL Dataset



(b) YaleB Dataset

Fig. 7: Variation of Objective Value with epoch for different ϵ_i on Face Datasets

comparing the convergence speed, we observe that our proposed **capeNMF** algorithm needs similar, if not lower, number of epochs to reach the optimal point.

Variation of Overall ϵ with ϵ_i on Face Datasets. In Figure 8, we show the variation of the final objective value, and the overall ϵ with ϵ_i for the **conventional – DP** algorithm and our proposed **capeNMF** algorithm on the two face datasets. As before, we show the objective value of the **non – private** algorithm as the baseline. We observe from the plots that if enough iterations are allowed, both the **conventional – DP** and **capeNMF** algorithms reach the same optimal point as the **non – private** algorithm, at the cost of higher overall ϵ since the overall ϵ depends on the number of iterations.

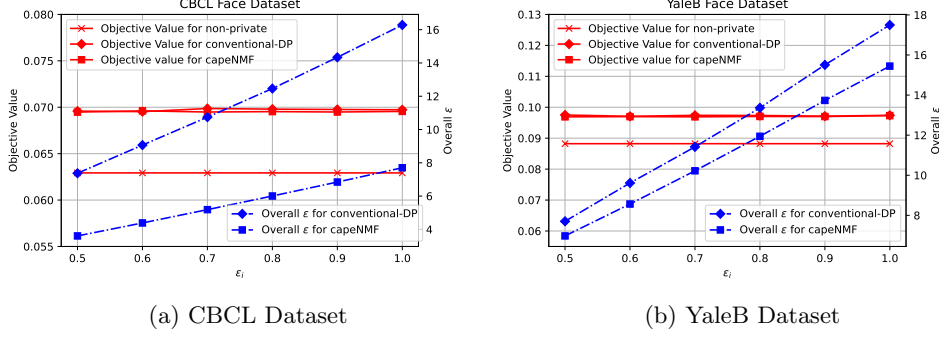


Fig. 8: Variation of Overall ϵ and Objective Value with ϵ_i on Face Datasets

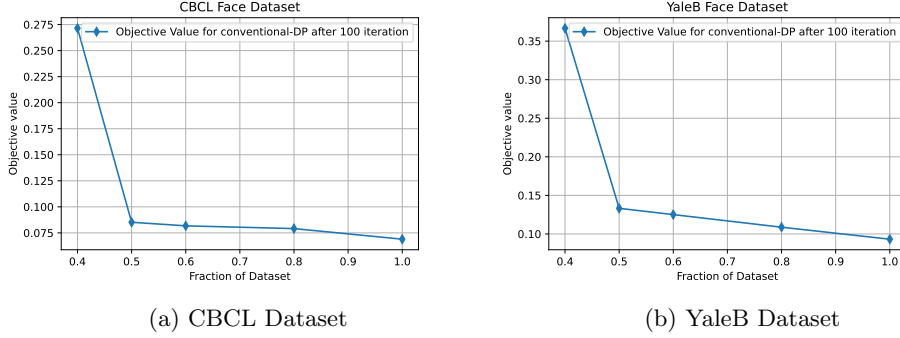


Fig. 9: Variation of Objective Value with dataset fraction on Face Datasets

We note that the proposed **capeNMF** algorithms typically requires a much smaller number of iterations to converge compared to **conventional – DP** due to the much lower noise (see Proposition 1 of Imtiaz et al. [8]). Therefore, the overall ϵ achieved by **capeNMF** is much smaller than that of **conventional – DP** algorithm. That is, our proposed **capeNMF** algorithm can provide a much better privacy guarantee compared to the **conventional – DP** approach.

Variation of Objective Value with Dataset Fraction on Face Datasets. In Figure 9, we show the variation of the final objective value with fraction of the face dataset used for a fixed $\epsilon_i = 0.5$. Similar to the text datasets, we observe that using the entire dataset produces the best results. In other words, our proposed algorithm performs better when more data samples are available.

Variation of Overall ϵ with Stopping Criteria on Face Datasets. We investigate the effect of stopping criteria on the overall ϵ of our proposed **capeNMF** algorithm. We set the same criteria for stopping the iterations at the aggregator: $|L_{\text{non-private}} - L_{\text{capeNMF}}| \leq \theta$, where $L_{\text{non-private}}$ and L_{capeNMF} are the final objective

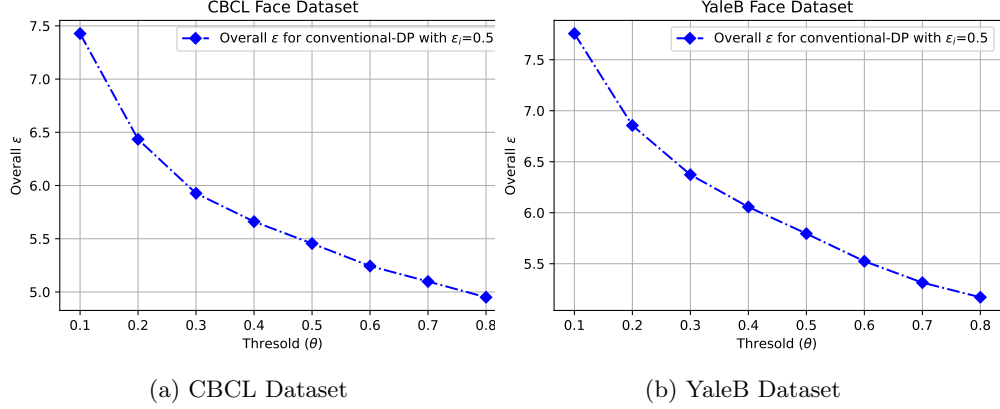


Fig. 10: Variation of Overall ϵ with Stopping Criteria on Face Datasets

values attained by the non – private algorithm, and the proposed **capeNMF** algorithm, respectively, for the face datasets. In Figure 10, we observe that for the two face datasets, the more closely we want to approximate the non-private algorithm’s objective value, the more iteration we would need, which inevitably results in higher overall ϵ . That is, we can reach a better optimal point at the cost of higher privacy risk.

5 Conclusion and Future Works

In this work, we proposed a novel privacy-preserving NMF algorithm (**capeNMF**) for estimating the differentially-private basis matrix $\mathbf{W}_{\text{private}}$ in a decentralized-data setting. We proposed to employ the CAPE scheme [8] to achieve utility close to the non-private setting. Additionally, we perform an analysis of our multi-round algorithm with Rényi Differential Privacy to achieve a tight characterization of the privacy spent. We performed experiments on six real datasets, and demonstrated that our proposed algorithm offered excellent performance, closely approximating that of the non-private algorithm. In most cases, our proposed **capeNMF** outperformed the conventional – DP approach by a significant margin. We presented plots that show the variation of the overall ϵ along with the objective value against the privacy parameter of each iteration, such that a data holder can select privacy budget per iteration, given the required utility level or privacy constraints. We also showed the effect of stopping criteria on the overall ϵ , which clearly indicated that our approach can reach the optimal point of the non – private algorithm given high-enough number of iterations, albeit at the cost of higher privacy risk. An interesting future work will be employing an optimal privacy mechanism (such as, the staircase mechanism [57]) in the decentralized-data setting.

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Appendix A Definitions and Theorems Related to Differential Privacy

We review some definitions and propositions that are necessary for our problem formulation according to Saha and Imtiaz [2].

Definition 1 ((ϵ, δ) -DP [3]). *An algorithm $f : \mathcal{D} \mapsto \mathcal{T}$ provides (ϵ, δ) -differential privacy ((ϵ, δ) -DP) if $\Pr(f(\mathbb{D}) \in \mathcal{S}) \leq \delta + e^\epsilon \Pr(f(\mathbb{D}') \in \mathcal{S})$ for all measurable $\mathcal{S} \subseteq \mathcal{T}$ and for all neighbouring datasets $\mathbb{D}, \mathbb{D}' \in \mathcal{D}$.*

Here, $\epsilon, \delta \geq 0$ are the privacy parameters, and determine how the algorithm will perform in providing privacy/utility. The parameter ϵ indicates how much the algorithm's output deviates in probability when we replace one single person's data with another. The parameter δ indicates the probability that the privacy mechanism fails to give the guarantee of ϵ . Intuitively, higher privacy results in poor utility. That is, smaller ϵ and δ guarantee more privacy, but lower utility. There are several mechanisms to implement differential privacy: Gaussian [3], Laplace mechanism [4], random sampling, and exponential mechanism [29] are well-known. Among the additive noise mechanisms, the noise standard deviation is scaled by the privacy budget and the sensitivity of the function.

Definition 2 ($\mathcal{L}_2$ sensitivity [3]). *The $\mathcal{L}_2$ -sensitivity of vector valued function $f(\mathbb{D})$ is $\Delta := \max_{\mathbb{D}, \mathbb{D}'} \|f(\mathbb{D}) - f(\mathbb{D}')\|_2$, where $\mathbb{D}$ and $\mathbb{D}'$ are neighbouring datasets.*

The $\mathcal{L}_2$ sensitivity of a function gives the upper bound of how much the function can change if one sample at the input is changed. Consequently, it dictates the amount of randomness/perturbation needed at the function's output to guarantee differential privacy. In other words, it captures the maximum change in the output by changing any one user in the worst-case scenario.

Definition 3 (Gaussian Mechanism [4]). *Let $f : \mathcal{D} \mapsto \mathcal{R}^D$ be an arbitrary function with $\mathcal{L}_2$ sensitivity Δ . The Gaussian mechanism with parameter τ adds noise from $\mathcal{N}(0, \tau^2)$ to each of the D entries of the output and satisfies (ϵ, δ) differential privacy for $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, if $\tau \geq \frac{\Delta}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}$.*

Here, (ϵ, δ) -differential privacy is guaranteed by adding noise drawn from $\mathcal{N}(0, \tau^2)$ distribution. Note that, there are an infinite number of combinations of (ϵ, δ) for a given τ^2 [2].

Definition 4 (Rényi Differential Privacy (RDP) [16]). *A randomized algorithm $f : \mathcal{D} \mapsto \mathcal{T}$ is (α, ϵ_r) -Rényi differentially private if, for any adjacent $\mathbb{D}, \mathbb{D}' \in \mathcal{D}$, the following holds: $D_\alpha(\mathcal{A}(\mathbb{D}) || \mathcal{A}(\mathbb{D}')) \leq \epsilon_r$. Here, $D_\alpha(P(x) || Q(x)) = \frac{1}{\alpha-1} \log \mathbb{E}_{x \sim Q} \left(\frac{P(x)}{Q(x)} \right)^\alpha$ and $P(x)$ and $Q(x)$ are probability density functions defined on $\mathcal{T}$.*

Proposition 1 (From RDP to DP [16]). *If f is an (α, ϵ_r) -RDP mechanism, it also satisfies $(\epsilon_r + \frac{\log 1/\delta}{\alpha-1}, \delta)$ -differential privacy for any $0 < \delta < 1$.*

Proposition 2 (Composition of RDP [16]). *Let $f_1 : \mathcal{D} \rightarrow \mathcal{R}_1$ be (α, ϵ_1) -RDP and $f_2 : \mathcal{R}_1 \times \mathcal{D} \rightarrow \mathcal{R}_2$ be (α, ϵ_2) -RDP, then the mechanism defined as (X_1, X_2) , where $X_1 \sim f_1(\mathbb{D})$ and $X_2 \sim f_2(X_1, \mathbb{D})$ satisfies $(\alpha, \epsilon_1 + \epsilon_2)$ -RDP.*

Proposition 3 (RDP and Gaussian Mechanism [16]). *If f has $\mathcal{L}_2$ sensitivity 1, then the Gaussian mechanism $\mathcal{G}_\sigma f(\mathbb{D}) = f(\mathbb{D}) + e$ where $e \sim \mathcal{N}(0, \sigma^2)$ satisfies $(\alpha, \frac{\alpha}{2\sigma^2})$ -RDP. Additionally, a composition of T such Gaussian mechanisms satisfies $(\alpha, \frac{\alpha T}{2\sigma^2})$ -RDP.*

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